Validation Execution Standard

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Canonical Definition

Validation Execution Standard (VEXS) defines the governance conditions under which an approved validation method is instantiated, performed, controlled, observed, documented, monitored, and completed against the established Validation Object, Validation Purpose, Validation Scope, Validation Criteria, qualified evidence basis, and reliability-qualified source structure.

Canonical Definition Extension

A validation method does not validate anything merely because it has been designed, selected, documented, or approved.

Validation requires execution.

Execution creates a separate governance problem.

A method may be appropriate while its execution is:

- incomplete,
- incorrectly sequenced,
- performed against the wrong object version,
- performed outside the approved scope,
- conducted under materially different conditions,
- based upon substituted evidence,
- affected by unauthorized deviations,
- interrupted,
- insufficiently documented,
- or impossible to reconstruct afterward.

VALIDOS® therefore separates:

Validation Method

from:

Validation Execution

Validation Method establishes **how validation should be performed.**

Validation Execution establishes **what validation was actually performed.**

VEXS governs the relationship between those two realities.

The standard establishes the conditions necessary to determine:

- whether the approved validation method was correctly instantiated,
- whether the correct Validation Object was examined,
- whether execution remained within approved Validation Scope,
- whether applicable Validation Criteria were actually evaluated,
- whether required evidence and sources were used appropriately,
- whether execution conditions were controlled,
- whether required procedural steps occurred,
- whether execution sequencing was preserved,
- whether deviations occurred,
- whether deviations were authorized,
- whether interruptions or failures affected execution,
- whether material substitutions occurred,
- whether execution remained sufficiently complete,
- whether execution records are reconstructable,
- and whether the completed execution is sufficiently conformant to proceed to Validation Determination.

VEXS does not determine whether the Validation Object satisfies the Validation Criteria.

It determines whether the **validation process itself was actually executed with sufficient conformity, completeness, control, and traceability for a later Validation Determination to be legitimate.**

Core Question

Was the approved validation process actually performed against the correct Validation Object, within the governed scope, according to the required method, criteria, evidence conditions, controls, and execution requirements?

Purpose

The purpose of Validation Execution Standard is to prevent validation conclusions from being derived merely from the existence of an approved validation methodology.

A theoretically excellent methodology provides little assurance if it was not executed as governed.

VEXS therefore creates a controlled bridge between:

Validation Design

and:

Validation Determination

The standard ensures that what was planned and what actually occurred can be compared, reconstructed, and governed independently.

Governance Objective

VEXS establishes a governed Validation Execution layer capable of determining whether the approved validation structure has been materially implemented in operational reality.

The standard seeks to ensure that:

- execution begins from an authorized validation basis,
- the correct Validation Object is used,
- the applicable object version or state is preserved,
- Validation Scope remains controlled,
- Validation Criteria remain connected to execution,
- the approved Validation Method is instantiated correctly,
- qualified evidence remains identifiable,
- source-reliability conditions remain visible,
- execution conditions are established,
- procedural steps remain traceable,
- deviations are detected,
- substitutions are controlled,
- interruptions are recorded,
- execution completeness can be assessed,
- execution conformity can be determined,
- and a reconstructable Validation Execution Record exists.

Operational Role

Validation Execution Standard serves as the **operational performance layer of the VALIDOS® validation lifecycle.**

The preceding Parent Standards establish:

Validation Object → Validation Purpose & Scope → Validation Criteria → Validation Method → Validation Evidence Fitness → Validation Source Reliability

VEXS then asks:

Was that governed validation design actually executed as required?

Its structural progression is:

Approved Validation Basis → Execution Authorization → Execution Instantiation → Controlled Performance → Execution Observation → Deviation Governance → Completeness Assessment → Execution Conformity → Validation Execution Record

The resulting execution record can then progress to Validation Determination.

Validation Execution Basis

Formal execution should begin from a sufficiently established Validation Execution Basis.

This may include:

- Validation Object reference,
- object version or state,
- Validation Purpose,
- Validation Scope,
- Validation Depth,
- Validation Criteria,
- approved Validation Method,
- qualified evidence basis,
- Source Reliability conditions,
- required execution controls,
- required participants,
- required systems or instruments,
- required operating conditions,
- and applicable restrictions.

The execution basis establishes the governed reference against which actual execution can later be assessed.

Execution Authorization

Validation execution should not begin merely because a method exists.

Where required by the validation context, execution should possess sufficient authorization to establish:

- that the validation is ready to proceed,
- that prerequisite governance conditions have been satisfied,
- that the approved method is the method intended for execution,
- that required resources are available,

- and that responsible execution roles are known.

Authorization requirements should remain proportionate to Validation Purpose, Depth, risk, and consequence.

Execution Instantiation

A general validation methodology must become a concrete execution instance.

Instantiation connects the approved method to:

- a specific Validation Object,
- a specific object version,
- a specific Validation Scope,
- applicable Validation Criteria,
- selected evidence,
- identified sources,
- execution participants,
- systems and instruments,
- time,
- environment,
- and other relevant operational conditions.

This prevents a generic method from being mistaken for evidence that validation actually occurred.

Validation Object Continuity

Execution must remain connected to the Validation Object established earlier in VALIDOS®.

VEXS therefore governs whether the object examined during execution corresponds to the object that was approved for validation.

Material differences involving:

- identity,
- version,
- configuration,
- state,
- boundaries,
- dependencies,
- or environment

may require execution reassessment or revalidation.

VALIDOS® establishes:

Validation of Object A cannot silently become validation of

Object B.

Validation Scope Conformity

Execution must remain within the governed Validation Scope.

VEXS determines whether execution:

- covered the required scope,
- exceeded the scope,
- omitted material scope elements,
- introduced unapproved exclusions,
- or operated under materially altered scope conditions.

A Validation Determination should not claim broader coverage than the execution actually performed.

Validation Criteria Execution

Applicable Validation Criteria must be connected to actual validation activity.

VEXS governs whether:

- required criteria were evaluated,
- evaluation activities are identifiable,
- omitted criteria are visible,
- conditional criteria were handled appropriately,
- and execution evidence can be connected to the relevant criteria.

VEXS does not determine whether each criterion was satisfied.

That belongs to Validation Determination.

Method Execution Conformity

The approved Validation Method creates the methodological reference for execution.

VEXS evaluates whether the actual process materially conformed to that method.

Relevant conditions may include:

- required steps,
- sequencing,
- sampling,
- testing procedures,
- analytical procedures,
- controls,

- observation requirements,
- repetition requirements,
- comparison procedures,
- simulation conditions,
- review procedures,
- and other method-specific requirements.

Planned Execution and Actual Execution

VALIDOS® distinguishes:

Planned Execution

from:

Actual Execution

The planned process describes what should occur.

The actual process records what did occur.

VEXS requires the relationship between the two to remain visible.

Where they differ materially, the difference becomes a governed Execution Deviation.

Execution Sequencing

Some validation activities depend upon order.

A later step may only be meaningful after an earlier condition has been established.

VEXS therefore governs sequencing where sequence affects methodological validity.

Incorrect sequencing may constitute:

- immaterial deviation,
- correctable deviation,
- material deviation,
- or execution-invalidating deviation

depending upon consequence.

Execution Conditions

Validation results may depend upon the conditions under which validation occurred.

Execution conditions may include:

- physical environment,
- digital environment,
- system configuration,
- network conditions,
- load,
- temperature,
- geographic context,
- population,
- operational state,
- human participation,
- available tools,
- simulation settings,
- or another materially relevant condition.

Required conditions should be recorded sufficiently to support later reconstruction.

Controlled Execution

Where the Validation Method requires controlled conditions, VEXS governs whether those conditions were sufficiently established and maintained.

Control may concern:

- environment,
- variables,
- access,
- data,
- configuration,
- instrumentation,
- participant behavior,
- or system state.

Loss of required control must remain visible.

Execution Evidence

Validation execution itself produces evidence.

This may include:

- test records,
- observations,
- measurements,
- system logs,
- analytical outputs,
- execution timestamps,
- operator records,
- instrument records,

- simulation results,
- exception records,
- screenshots,
- audit trails,
- or other execution artifacts.

These artifacts form part of the Validation Execution Record.

Evidence Continuity During Execution

Evidence qualified earlier in VALIDOS® should remain connected to its governed identity and source conditions during execution.

VEXS prevents:

- evidence substitution without record,
- mixing evidence versions,
- using expired evidence without governance,
- removing evidence limitations,
- ignoring Source Reliability restrictions,
- or silently replacing approved evidence with alternative material.

Source Condition Continuity

Source Reliability conditions established under VSRS must remain visible during execution.

For example:

If a source is reliable only:

- within a defined role,
- under specific conditions,
- with independent corroboration,
- within a particular version,
- or subject to a restriction,

execution must preserve those conditions.

VEXS does not reassess Source Reliability unless material change requires reassessment.

It ensures that source conditions are respected during execution.

Execution Participant Governance

Validation may involve:

- validators,
- operators,
- observers,
- reviewers,
- experts,
- automated systems,
- AI agents,
- autonomous systems,
- or other participants.

VEXS requires material execution roles to be sufficiently identifiable where their actions affect validation reconstruction or accountability.

Human Execution

Where humans perform validation activities, execution governance may include:

- assigned role,
- required instructions,
- procedural conformity,
- relevant competence requirements,
- intervention records,
- manual overrides,
- observations,
- and deviations.

Human participation should remain distinguishable from automated execution where material.

Automated Execution

Validation may be executed partially or entirely by automated systems.

Automated execution may involve:

- testing pipelines,
- monitoring systems,
- analytical software,
- simulation systems,
- AI systems,
- autonomous agents,
- or machine-controlled procedures.

VEXS applies the same fundamental question:

Did the governed validation process actually occur as required?

Automation does not remove the requirement for execution traceability.

Hybrid Execution

Many validation processes combine human and automated activity.

VEXS governs the interaction points where:

- humans initiate automated processes,
- automated systems generate findings,
- humans interpret outputs,
- systems escalate exceptions,
- humans override automation,
- or multiple systems exchange execution responsibility.

Material handoffs should remain reconstructable.

Execution Deviation

An **Execution Deviation** occurs when actual execution differs from the approved execution basis.

Examples include:

- skipped step,
- changed sequence,
- substituted evidence,
- changed instrument,
- altered configuration,
- reduced sample,
- unavailable source,
- environmental variation,
- manual intervention,
- unexpected system behavior,
- or unplanned interruption.

Deviation is not automatically execution failure.

It is a governance event.

Deviation Materiality

VEXS distinguishes deviations according to their potential effect.

Possible classes may include:

Immaterial Deviation

Minor Deviation

Material Deviation

Critical Deviation

Execution-Invalidating Deviation

Classification should depend upon the Validation Purpose, Criteria, Method, Depth, and consequence.

Authorized Deviation

Some deviations may be anticipated or legitimately approved.

An Authorized Deviation should establish:

- what changed,
- why it changed,
- who or what authorized it,
- when authorization occurred,
- which validation elements were affected,
- and whether compensating controls were required.

Authorization does not automatically make a deviation methodologically harmless.

Its effect must still be assessed.

Unauthorized Deviation

An Unauthorized Deviation occurs where actual execution departs materially from the approved basis without sufficient authorization.

Such deviation may require:

- correction,
- repetition,
- restricted use,
- additional validation,
- escalation,
- or execution invalidation.

Execution Substitution

Substitution occurs where a planned validation component is replaced during execution.

Substitution may involve:

- evidence,
- source,
- instrument,
- method step,
- participant,
- dataset,
- model,
- system,
- environment,
- or control.

Material substitutions must remain explicit.

Equivalent appearance does not establish methodological equivalence.

Execution Interruption

Execution may be interrupted because of:

- system failure,
- loss of evidence,
- infrastructure outage,
- human intervention,
- safety condition,
- environmental change,
- resource failure,
- or another event.

VEXS governs whether execution can:

- resume,
- restart,
- continue conditionally,
- require partial repetition,
- or require complete re-execution.

Execution Failure

An Execution Failure occurs where required validation activity cannot be performed with sufficient methodological integrity.

Failure may affect:

- one criterion,
- one execution stage,
- one evidence source,
- one test,
- or the complete validation execution.

Execution failure must not automatically be converted into failure
of the Validation Object.

VALIDOS® establishes:

Failure to Validate ≠ Validation Failure of the Object

This distinction is fundamental.

Execution Completeness

Execution completeness asks whether all materially required
validation activities were actually performed.

Completeness may be:

Complete

Conditionally Complete

Partially Complete

Incomplete

or:

Completeness Undetermined

Completeness concerns execution coverage.

It does not determine criterion satisfaction.

Execution Conformity

Execution Conformity asks whether the completed validation process
sufficiently corresponds to the approved Validation Execution Basis.

Possible states may include:

Execution Conformant

The material validation activities were performed in sufficient
conformity with the approved execution basis.

Execution Conditionally Conformant

Execution may support subsequent determination only under explicit
conditions, limitations, or restrictions.

Execution Conformity Undetermined

Available execution information is insufficient to establish conformity.

Execution Non-Conformant

Material execution requirements were not sufficiently followed.

Execution Invalid

Execution contains critical conditions preventing legitimate use for the intended Validation Determination.

These states apply to the validation process.

They do not determine whether the Validation Object is valid.

Execution Conformity Is Not Criterion Satisfaction

VALIDOS® establishes:

Execution Conformant ≠ Validation Criteria Satisfied

A perfectly executed validation may demonstrate that the Validation Object fails.

That is a valid outcome.

Likewise, an object that might actually satisfy the criteria cannot legitimately receive a governed positive Validation Determination from materially invalid execution.

Execution Completeness Is Not Validation Success

A complete validation means the required validation activities were performed.

It does not mean the object passed them.

Therefore:

Complete Execution ≠ Successful Validation Outcome

This preserves the separation between process quality and object validity.

Execution Reconstruction

A validation process should be sufficiently reconstructable to determine:

- what was performed,
- against which object,
- by whom or by what,
- using which method,
- using which evidence,

- under which conditions,
- in which sequence,
- with which deviations,
- and with which resulting execution artifacts.

The required reconstruction depth should remain proportionate to validation consequence.

Execution Traceability

VEXS establishes traceability between:

Validation Object

Validation Purpose & Scope

Validation Criteria

Validation Method

Validation Evidence

Validation Sources

Execution Activities

Execution Deviations

and:

Execution Outputs

This creates the operational chain required for defensible Validation Determination.

Execution Record

A formal **Validation Execution Record** should preserve, where applicable:

- Validation Object reference,
- object version or state,
- Validation Purpose,
- Validation Scope,
- Validation Depth,
- Validation Criteria references,
- Validation Method reference,
- evidence references,
- source references,
- Source Reliability conditions,
- execution authorization,
- execution participants,

- execution environment,
- execution sequence,
- performed activities,
- timestamps,
- execution outputs,
- deviations,
- substitutions,
- interruptions,
- failures,
- corrective actions,
- execution completeness,
- execution conformity,
- and sufficient auditability.

Execution Repetition

A validation activity may require repetition where:

- execution failed,
- critical deviation occurred,
- required conditions were not maintained,
- evidence was materially substituted,
- system configuration changed,
- or execution records are insufficient.

Repetition should preserve the relationship between the original and repeated execution.

Partial Re-Execution

Not every execution issue requires repeating the complete validation.

Where methodologically defensible, VEXS may permit partial re-execution of affected validation activities.

The unaffected portions must remain demonstrably separable from the affected portions.

Execution Versioning

Repeated or materially changed validation executions should remain distinguishable.

Possible representations may include:

Execution 1.0

Execution 1.1

Execution 2.0

or another governed versioning structure.

Historical execution records should not be silently overwritten.

Execution Change Governance

Material changes occurring during execution may trigger:

- continuation,
- controlled deviation,
- partial restart,
- complete restart,
- scope reassessment,
- method reassessment,
- evidence reassessment,
- source reassessment,
- or revalidation.

The appropriate response depends upon what changed and how materially it affects validation.

Execution Readiness for Determination

A completed execution may progress to Validation Determination when sufficient information exists to determine:

- what was executed,
- whether the correct object was used,
- whether scope was respected,
- whether applicable criteria were addressed,
- whether the approved method was materially followed,
- whether evidence and source conditions were preserved,
- whether material deviations are understood,
- whether execution is sufficiently complete,
- whether execution is sufficiently conformant,
- and whether the execution record is reconstructable.

Determination Block

Where execution is materially insufficient, VEXS may establish a **Determination Block.**

A Determination Block prevents a final Validation Determination from being represented as methodologically established until the blocking execution condition is resolved.

Possible causes include:

- critical missing execution activity,

- wrong Validation Object,
- critical scope deviation,
- invalid method substitution,
- irrecoverable evidence discontinuity,
- critical execution failure,
- or insufficient execution traceability.

Governance Boundary

Validation Execution Standard begins when the governed validation basis is sufficiently established for operational execution.

It ends when actual execution has been performed, recorded, assessed for completeness and conformity, and determined sufficiently ready—or not ready—to enter Validation Determination.

VEXS does not determine:

- whether the Validation Criteria were ultimately satisfied,
- the final Validation Determination,
- the formal Validation State,
- reliance authorization,
- validation expiration,
- or future revalidation requirements.

These responsibilities belong to subsequent VALIDOS® Parent Standards.

Minimum Governance Requirements

A conforming implementation of VEXS should establish:

- 1. an identifiable Validation Execution Basis,
- 2. the Validation Object and applicable version or state,
- 3. execution authorization where required,
- 4. applicable Validation Scope and Depth,
- 5. Validation Criteria references,
- 6. approved Validation Method reference,
- 7. qualified evidence and source-condition references,
- 8. identifiable execution activities,
- 9. relevant execution participants or systems,

10. material execution conditions, 11. execution sequencing where required, 12. deviation and substitution governance, 13. interruption and failure governance, 14. Execution Completeness assessment, 15. Execution Conformity assessment, 16. a reconstructable Validation Execution Record, 17. and a determination of readiness for Validation Determination.

Governance Outputs

VEXS may produce:

- Validation Execution Basis,
- Validation Execution Authorization,
- Validation Execution Instance,
- Validation Execution Plan Reference,
- Validation Activity Record,
- Validation Criteria Execution Map,
- Validation Evidence Use Record,
- Source Condition Continuity Record,
- Execution Environment Record,
- Execution Sequence Record,
- Execution Observation Record,
- Execution Deviation Record,
- Execution Substitution Record,
- Execution Interruption Record,
- Execution Failure Record,
- Corrective Execution Record,
- Execution Completeness Determination,
- Execution Conformity Determination,
- Validation Execution Record,
- Determination Readiness Status,
- Determination Block,
- and Execution Traceability Record.

Operational Applications

Validation Execution Standard may be applied across:

- artificial intelligence,
- machine learning,
- autonomous systems,
- model validation,
- AI safety testing,
- software validation,
- data validation,
- predictions,
- simulations,
- scientific validation,
- financial services,
- healthcare,
- industrial operations,
- manufacturing,
- regulatory compliance,
- critical infrastructure,
- cybersecurity,
- digital identities,
- operational governance,
- public-sector systems,
- and any governed environment requiring demonstrable validation execution.

Relationship to Other OOF Architectures

Validation Execution Standard interoperates with:

- **GOA™**
- **OBIDENITY®**
- **INTEGROS®**
- **ORA™**
- **AGA™**
- **AIG®**
- **CLIA®**
- **MGIA™**
- **ASGA™**
- **RIS™**

by providing the canonical validation-governance methodology for determining whether an approved validation structure was actually instantiated and performed with sufficient conformity, completeness, control, continuity, and traceability.

GOA™ may provide complementary authority and delegation conditions.

OBIDENITY® may provide complementary identity, origin, ownership, provenance, and continuity relationships.

INTEGROS® may provide complementary integrity conditions affecting execution records and processes.

ORA™ may provide complementary operational-reality information.

AGA™ may provide complementary accountability relationships for execution roles and decisions.

AIG®, CLIA®, MGIA™, and **ASGA™** may establish specialized execution conditions for AI behavior, cognition, memory, and autonomous systems.

RIS™ may inform the execution depth and control strength required according to risk and consequence.

VEXS does not duplicate these architectures.

It consumes relevant governed information where necessary and determines what that information means specifically for **Validation Execution.**

Foundational Principle

A validation method has no operational validity until its required validation activities are actually performed.

Canonical Closing Statement

Validation Execution Standard establishes the operational performance layer of VALIDOS® — Validation Governance Architecture by governing the transition from approved validation design to demonstrable validation activity. Through execution authorization, instantiation, object continuity, scope conformity, criteria execution, method conformity, evidence and source continuity, execution conditions, sequencing, deviations, substitutions, interruptions, failures, completeness, conformity, reconstruction, and traceability, VEXS ensures that no Validation Determination derives legitimacy merely from what validation was intended to do, but from what the governed validation process can demonstrate was actually performed.

Validation Execution Standard

Architecture: VALIDOS® — Validation Governance Architecture
Category: AI & Interpretation · **Subcategory:** Validation Governance

Canonical Definition: Validation Execution Standard (VEXS) defines the governance conditions under which an approved validation method is instantiated, performed, controlled, observed, documented, monitored, and completed against the established Validation Object, Validation Purpose, Validation Scope, Validation Criteria, qualified evidence basis, and reliability-qualified source structure.

Governed Space: Validation Execution

A validation method has no operational validity until its required validation activities are actually performed.

→ View Standard

Module Architecture

→ Validation Execution Basis & Authorization Module (VEBAM)

→ Validation Execution Instantiation & Control Module (VEICM)

→ Validation Execution Observation & Traceability Module (VEOTM)

→ Validation Execution Deviation & Exception Governance Module (VEDEGM)

→ Validation Execution Completeness & Conformity Module (VECCM)

Parent Resources

→ VALIDOS® Architecture Map

→ VALIDOS® Complete Standards & Modules Index

→ VALIDOS® — Four-Layer Validation Protocol™

Standards Compatibility

→ Validation Method Standard
Operational compatibility within the governed validation lifecycle.

→ Validation Evidence Fitness Standard
Operational compatibility within the governed validation lifecycle.

→ Validation Source Reliability Standard
Operational compatibility within the governed validation lifecycle.

→ Validation Determination Standard
Operational compatibility within the governed validation lifecycle.

→ Validation State Governance Standard
Operational compatibility within the governed validation lifecycle.

Related Documents

→ About Validation Execution Standard

→ VALIDOS® Validation Governance Architecture — Validation Governance Layer

OOF® MIP® Protection Notice

Canonical Source

https://originopenfoundation.org/